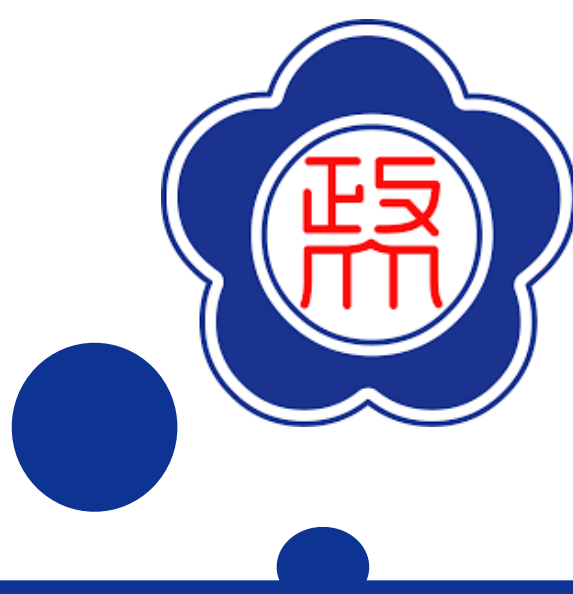




The Effect of Words' Visual Complexity Distribution on Saccade Targeting in Reading Chinese Sentences

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Introduction

Several studies have examined the influence of orthographic language features on oculomotor control and saccade programming in alphabetic languages, such as *word length* and *inter-word spacing* (Rayner, 1979; Rayner, Fischer, & Pollatsek, 1998; McConkie, Kerr, Reddix, & Zola, 1988). Rayner (1979) was the first author to propose the existence of a *saccade strategy*, making reference to the oculomotor mechanism responsible for deciding where to send the eyes next; this statement was inferred from the proportion of *Initial Landing Position* (ILP) of inter-word saccades in words of different lengths. Rayner found that the ILP within a word was modulated by its length and when making inter-word forward saccades the readers' eyes tended to land half way between the end and the middle of the word; that is to say, word length information is actively used by English readers for moving their eyes along the text and choosing saccade targets. From these observations it was suggested the existence of a *Preferred Viewing Location* (PVL) near the word center, result that was taken as evidence supporting a word-based saccade strategy in reading English, finding that was later replicated in reading other languages such as German, Hebrew and Uighur (Deutsch & Rayner, 1999; Nuthmann & Kliegl, 2013; Yan et al., 2014). The validation of the PVL phenomenon across alphabetic languages indicates that inter-word spaces are salient visual cues actively used in deciding where to send the eyes next. The present study attempts to address the question about the relative weight of the visual patterns of traditional Chinese characters upon the native Chinese readers' *saccade strategy*; for doing it so and in view of the results of a flat PVL within 2-character words when reading natural sentences (Yang, 1997; Yang & George, 1999), we devised an index that reflects the *Visual Complexity Distribution* (VCD) within 2-character Chinese words and tested its influence on different eye movement measures.

Materials

Several findings suggest that saccades are sent towards word units and not towards individual characters neither towards sub-lexical units when reading Chinese (Yang & George, 1999; Yan, Kliegl, Richter, Nuthmann, & Shu, 2010; Yan, Kliegl, Shu, Pan, & Zhou, 2010; Zang, Meng, Liang, Bai, & Yan, 2013); thus a new index that captures the *Visual Complexity Distribution* (VCD) along the 2-character words' area was devised and tested experimentally. We computed the VCD index of 2-character words available in the *Academia Sinica Chinese Corpus* by displaying each word in a monitor screen using *kai* font with a size of 32² pixels in black color on white background; we then used a script that allowed us to retrieve the color values of each pixel, make a count of the pixels filled by the characters' strokes within the word divided horizontally into 8 bins and calculated the slope of the regression line along these bins as shown in Figure 1.

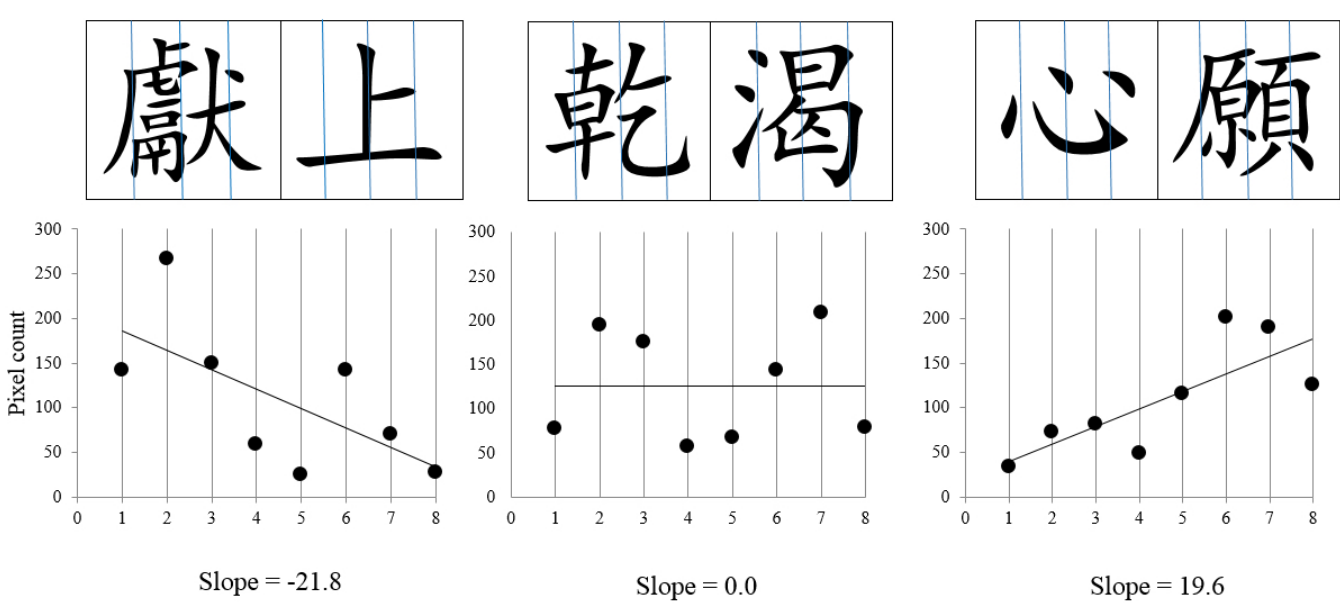


Figure 1: Visualization of the words' VCD Index computation

Participants

First-year psychology students from National Chengchi University participated in the experiment ($n = 42$; *age range* = 19 – 21); all participants were Taiwanese, native Chinese speakers, had normal or corrected-to-normal vision and were naive concerning the purpose of the experiment.

Apparatus and Methods

A SR-Research EyeLink 1000 eye tracking system was used to monitor and record the participants' eye movement at a rate of 1000 Hz. Viewing was binocular but only the participant's dominant eye was recorded; participants read single-line Chinese sentences from a computer monitor (resolution: 1024 × 768 pixels), the characters were displayed in black color in light gray background and the reading direction of the sentence was horizontal from left to right. The characters were displayed in *kai* font type in a size of 32² pixels with a space of 4 pixels between each character; in this way, at a viewing distance of 70 cm, one character with the space before it subtended approximately 1° of visual angle. First, participants read 8 practice sentences and 2 comprehension questions for getting to familiarize with the experimental procedure; after solving the doubts about the experiment, participants started to read the experimental sentences. Each participant read 138 experimental sentences in 3 blocks and the first 4 sentences of each block were fillers whose data was excluded from subsequent analysis. Sentence were presented once to each participant with the three conditions counterbalanced over participants following a Latin square design. After the eye movement data was collected, the fixations were determined with an algorithm for binocular saccade detection (Engbert & Kliegl, 2003).

Variables and Data Analyses

The eye movement measures that were thought to be relevant to our research question were: *a) First Fixation Duration* (FFD); *b) Single Fixation Duration* (SFD), both measures reflecting a relatively early stage of lexical access and fovea processing; and *c) Gaze duration* (GD), measure that reflects the integration of visual and semantic features of the word. Additionally and in order to test the influence of the words' VCD index on saccade targeting we were interested on the ILP of first and single fixations (i.e. PVL FF and PVL SF respectively). Statistical inferences about the VCD index's effects on these eye movement measures are based on Linear Mixed Models (LMMs) and the experimental effects were estimated by a contrast matrix between pair of conditions. Additionally and given the large influence that have been attributed to the *launch site* on ILP, we considered the launch site of first-forward saccades as a covariate term or confounding factor; this allowed us to measure its main effect and partial it out in order to estimate the experimental effects on the ILP of first and single fixations. All of the statistical procedures were done using the statistical software R, version 3.1.1 (Team et al., 2013), multivariate

analyses were carried out using the *lmer* function of the *lme4* package (Bates, 2010) and the figures were created with the *ggplot* package for data visualization (Wickham, 2016). Finally, we used the *remef* package (Hohenstein & Kliegl, 2015) for removing the random factors variance and fixed effects from the dependent variable of each model.

Results

The descriptive statistics are graphically presented in Figure 2 for each of the eye movement variables mentioned above, in this figure each bar stands for a particular VCD condition and the vertical line on it denotes 1.5 times the standard error of the mean. Table 1 presents the fixed effects of the planned contrasts for two models along the different eye movement measures; the *main model* (upper part), contrasted the responses on LB and NB target words against RB words (i.e. LB - RB and NB - RB); in the *alternative model* (bottom part) the observations on LB and RB target words are contrasted with those on NB target words (i.e. LB - NB).

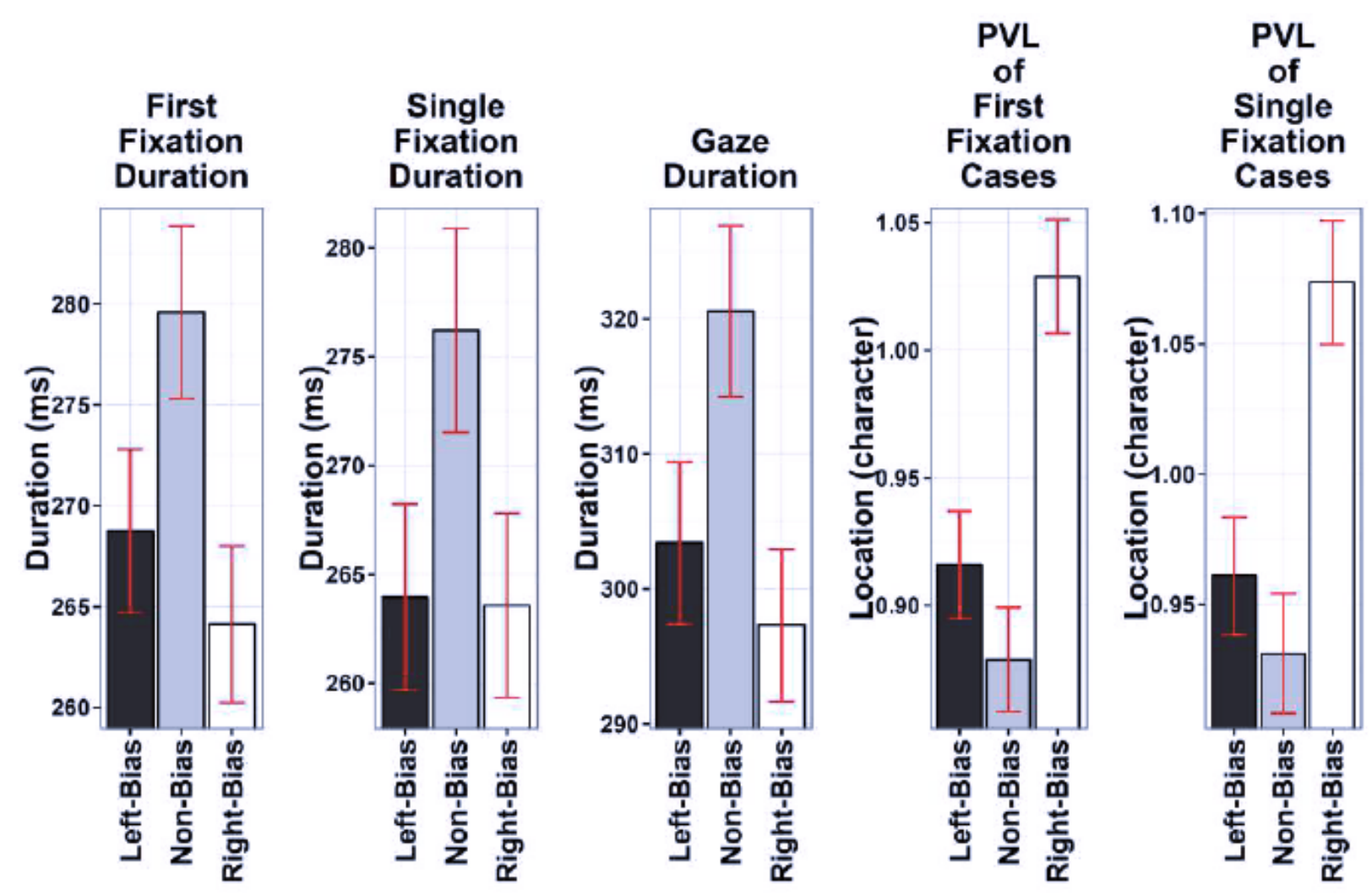


Figure 2: Mean for each eye movement measure by VCD condition

Figure 2 shows that the word identification or duration measures (i.e. FFD and SFD) as well as the total fixation time (i.e. GD), were significantly larger for the NB condition than for the LB and RB conditions; reaching significant levels for the contrast LB - NB (FFD: $t = -2.14$; SFD: $t = -2.41$; GD: $t = -2.32$; all $p < 0.05$) as well as for the contrast NB - RB (FFD: $t = 3.06$; SFD: $t = 2.52$; GD: $t = 3.00$; all $p < 0.05$). Instead, the contrast between LB - RB conditions didn't reach significant levels (FFD: $t = 0.92$; SFD: $t = 0.13$; GD: $t = 0.69$; all $p > 0.3$). Additionally and for the position measures it was found that the PVL of the RB condition was relatively further into the word in contrast to LB and NB conditions for both sets of initial fixations, reaching significant levels in the LB - RB contrast (PVL FF: $t = -4.81$; PVL SF: $t = -4.73$; all $p < 0.01$), as well as in the NB - RB contrast (PVL FF: $t = -6.40$; PVL SF: $t = -5.94$; all $p < 0.01$). Furthermore, the PVL of LB and NB conditions was found to be close from each other, as suggested by the lack of statistical significance in the contrast LB - NB (PVL FF: $t = 1.60$; PVL SF: $t = 1.25$; all $p > 0.1$).

Table 1: Estimated effects on eye movement measures using LMM

Model	Effects	FFD	SFD	GD	PVL FF	PVL SF
Main	LB - RB	0.02 (0.02)	0.002 (0.02)	0.02 (0.02)	-0.11 (0.02) **	-0.11 (0.02) **
	NB - RB	0.05 (0.02) **	0.05 (0.02) *	0.07 (0.02) **	-0.15 (0.02) **	-0.14 (0.02) **
	Launch Site	-0.004 (0.01)	0.01 (0.01)	-0.03 (0.01) **	0.36 (0.01) **	0.34 (0.01) **
	Intercept	5.56 (0.02) **	5.55 (0.02) **	5.66 (0.03) **	0.94 (0.04) **	0.99 (0.03) **
Alternative	LB - NB	-0.04 (0.02) *	-0.04 (0.02) *	-0.05 (0.02) *	0.04 (0.02)	0.03 (0.02)
	Observations	3573	2851	3464	3573	2851

The standard errors of estimated effects are reported in parenthesis for different eye movement measures. Signif. codes: 0.001 *** 0.05 ** *

Conclusions

- The findings on this experiment provide some evidence suggesting the VCD index's effects on saccade targeting mechanism and oculomotor control in reading natural traditional Chinese sentences.
- The PVL FF and SF cases was further into the word for the RB condition in comparison to that of the NB and LB conditions, thus evidencing that information about the luminance patterns of words located in parafoveal vision is used when specifying saccade targets. The influence of the words' VCD on saccade targeting mechanism might take place only after readers have isolated the currently fixated word from the string of characters, which implies the awareness about the left boundary position of word $n + 1$. Thus, when LB and NB words were located in parafoveal vision a saccade strategy based on the awareness about the location of a semantic boundary is more likely to be deployed, resulting in comparable PVL for those words. Instead, the VCD index of RB words was perceived for the readers as an additional cue about how the visual components are distributed along the area of the word to be fixated next, which shifted the PVL rightwards.
- The LMMs results demonstrated the relatively large effect of launch site on the saccade targeting mechanism as indicated by the models on PVL FF and SF cases; in consequence, when the launch site of the incoming saccade was closer to the word's beginning, the eyes were sent further into the word and allowed a larger preprocessing of words located in parafoveal vision as reflected by its significant effect on the GD model.
- The analyses on the duration measures indicated that readers of Chinese require more time to integrate visual and semantic features of NB words in contrast to LB and RB words, as suggested by the early processing measures (i.e. FFD and SFD) regardless of having controlled both character and word frequency, as well as the overall stroke count across conditions. In addition, the integration of visual and semantic features for the NB words was to some extent delayed as suggested by relatively larger GD times.

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References

Bates, D. M. (2010). *lme4: Mixed-effects modeling with r*. Berlin.

Cai, D., Chi, C.-F., & You, M. (2001). The legibility threshold of chinese characters in three-type styles. *International Journal of Industrial Ergonomics*, 27(1), 9–17.

Deutsch, A., & Rayner, K. (1999). Initial fixation location effects in reading hebrew words. *Language and Cognitive Processes*, 14(4), 393–421.

Engbert, R., & Kliegl, R. (2003). Microsaccades uncover the orientation of covert attention. *Vision research*, 43(9), 1035–1045.

Hohenstein, S., & Kliegl, R. (2015). *remef: Remove partial effects (r package version 1.0. 6.9000)[computer software]*.

McConkie, G. W., Kerr, P. W., Reddix, M. D., & Zola, D. (1988). Eye movement control during reading: I. the location of initial eye fixations on words. *Vision research*, 28(10), 1107–1118.

Nuthmann, A., & Kliegl, R. (2013). Preferred viewing locations: A validation and an extension. *The Mind Research Repository (beta)*(1).

Rayner, K. (1979). Eye guidance in reading: Fixation locations within words. *Perception*, 8(1), 21–30.

Rayner, K., Fischer, M. H., & Pollatsek, A. (1998). Unspaced text interferes with both word identification and eye movement control. *Vision Research*, 38(8), 1129–1144.

Rayner, K., Pollatsek, A., Ashby, J., & Clifton Jr, C. (2012). *Psychology of reading*. Psychology Press.

Reichle, E. D., Pollatsek, A., Fisher, D. L., & Rayner, K. (1998). Toward a model of eye movement control in reading. *Psychological review*, 105(1), 125.

Team, R. C., et al. (2013). R: A language and environment for statistical computing.

Wickham, H. (2016). *ggplot2: elegant graphics for data analysis*. Springer.

Yan, M., Kliegl, R., Richter, E. M., Nuthmann, A., & Shu, H. (2010). Flexible saccade-target selection in chinese reading. *The Quarterly Journal of Experimental Psychology*, 63(4), 705–725.

Yan, M., Kliegl, R., Shu, H., Pan, J., & Zhou, X. (2010). Parafoveal load of word n+ 1 modulates pre-processing effectiveness of word n+ 2 in chinese reading. *Journal of Experimental Psychology: Human Perception and Performance*, 36(6), 1669.

Yan, M., Zhou, W., Shu, H., Yusupu, R., Miao, D., Krügel, A., & Kliegl, R. (2014). Eye movements guided by morphological structure: Evidence from the uighur language. *Cognition*, 132(2), 181–215.

Yang, H.-M. (1997). Optimal viewing position in chinese reading. *Bulletin of Special Education and Rehabilitation*(5), 37–59.

Yang, H.-M., & George, M. (1999). Reading chinese: Some basic eye-movement characteristics. *Reading Chinese script*, 207–222.

Zang, C., Meng, H., Liang, F., Bai, X., & Yan, G. (2013). Is there a vertical component to saccade targeting in chinese reading? *Journal of Research in Reading*, 36, S78–S93.